Precision, Quantization, and Numerical Correctness

Deep Dive into DeepSeek — Chapter 32

Yin Yang

Foundation Books Project

The one rule of this chapter

A payload is not a number. A payload plus its scale, accumulator, reference, metric, and tolerance is a number.

- Quantization replaces tensor values with compact approximations chosen from a smaller representable set.
- That can save bytes and accelerate arithmetic — but rounding and clipping introduce error, and identical payload bytes under different scales are different tensors.
- Whether the resulting error is acceptable is a separate question from whether the payload is small.

The one rule of this chapter

A payload is not a number. A payload plus its scale, accumulator, reference, metric, and tolerance is a number.

- Quantization replaces tensor values with compact approximations chosen from a smaller representable set.
- That can save bytes and accelerate arithmetic — but rounding and clipping introduce error, and identical payload bytes under different scales are different tensors.
- Whether the resulting error is acceptable is a separate question from whether the payload is small.

Goal: follow one small projection from a BF16 reference through FP8 and FP4-style paths, and see exactly where rounding enters, where it compounds, and when it flips a decision instead of just nudging a number.

Roadmap

1. The correctness contract

Roadmap

1. The correctness contract
2. From BF16 down to FP4

Roadmap

1. The correctness contract
2. From BF16 down to FP4
3. Granularity, accumulation, and propagation

Roadmap

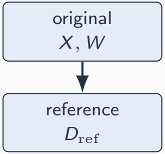
1. The correctness contract
2. From BF16 down to FP4
3. Granularity, accumulation, and propagation
4. Where numerical error becomes a decision

Roadmap

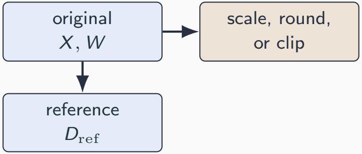
1. The correctness contract
2. From BF16 down to FP4
3. Granularity, accumulation, and propagation
4. Where numerical error becomes a decision
5. Testing, training vs. inference, and what carries forward

The correctness contract

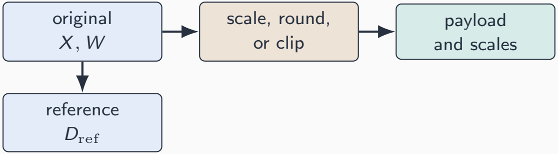
Encode, store, reconstruct, compare



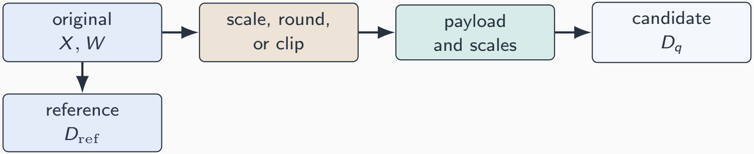
Encode, store, reconstruct, compare



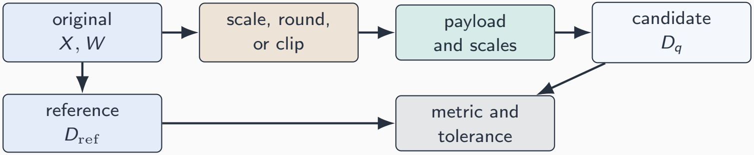
Encode, store, reconstruct, compare



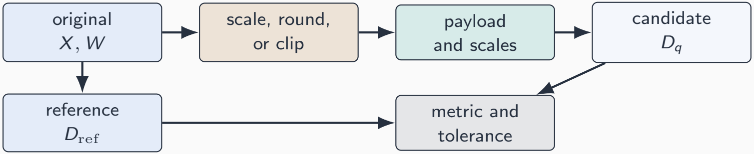
Encode, store, reconstruct, compare



Encode, store, reconstruct, compare



Encode, store, reconstruct, compare



A named reference, metric, and tolerance decide acceptance — not the compactness of the encoded path.

The running projection, carried all chapter

$$D_{\mathrm{ref}} = C + XW^\top, \qquad D_q = C + \widetilde{X}\widetilde{W}^\top,$$

with $X \in \mathbb{R}^{4\times 8}$, $W \in \mathbb{R}^{6\times 8}$, $C \in \mathbb{R}^{4\times 6}$.

- The quantized path stores payloads P_X, P_W with interpretation metadata m_X, m_W, then reconstructs $\widetilde{X}, \widetilde{W}$.

The running projection, carried all chapter

$$D_{\text{ref}} = C + XW^{\top}, \qquad D_q = C + \widetilde{X}\widetilde{W}^{\top},$$

with $X \in \mathbb{R}^{4\times 8}$, $W \in \mathbb{R}^{6\times 8}$, $C \in \mathbb{R}^{4\times 6}$.

- The quantized path stores payloads P_X, P_W with interpretation metadata m_X, m_W, then reconstructs $\widetilde{X}, \widetilde{W}$.
- **Numerical correctness contract** (Def.): the tuple (format, scale rule, accumulator, reference, metric, tolerance, shape family, environment).

The running projection, carried all chapter

$$D_{\mathrm{ref}} = C + XW^{\top}, \qquad D_q = C + \widetilde{X}\widetilde{W}^{\top},$$

with $X \in \mathbb{R}^{4\times 8}$, $W \in \mathbb{R}^{6\times 8}$, $C \in \mathbb{R}^{4\times 6}$.

- The quantized path stores payloads P_X, P_W with interpretation metadata m_X, m_W, then reconstructs $\widetilde{X}, \widetilde{W}$.
- **Numerical correctness contract** (Def.): the tuple (format, scale rule, accumulator, reference, metric, tolerance, shape family, environment).

Exact equality is often the wrong oracle: a perturbation that is harmless for a wide-margin projection can flip a near-tied attention or routing decision.

From BF16 down to FP4

BF16: a recorded procedure, not just a dtype name

Tensor	Entries	BF16 bytes
X	32	64

BF16: a recorded procedure, not just a dtype name

Tensor	Entries	BF16 bytes
X	32	64
W	48	96

BF16: a recorded procedure, not just a dtype name

Tensor	Entries	BF16 bytes
X	32	64
W	48	96
C, D_{ref}	24 each	48 each

BF16: a recorded procedure, not just a dtype name

Tensor	Entries	BF16 bytes
X	32	64
W	48	96
C, D_{ref}	24 each	48 each

- A reference computation names shapes, storage dtypes, accumulator convention, implementation, and seed — *then* a tolerance can name it.
- Accumulator (partial-sum path) and output dtype (stored result) are independent fields; FP8 input storage leaves the accumulator open, and BF16 output storage leaves the product path open.

BF16: a recorded procedure, not just a dtype name

Tensor	Entries	BF16 bytes
X	32	64
W	48	96
C, D_{ref}	24 each	48 each

- A reference computation names shapes, storage dtypes, accumulator convention, implementation, and seed — *then* a tolerance can name it.
- Accumulator (partial-sum path) and output dtype (stored result) are independent fields; FP8 input storage leaves the accumulator open, and BF16 output storage leaves the product path open.

If X, W, C, D_{ref} are all BF16-stored, the byte counts above are exactly what you can inspect — performance still needs a timed run.

FP8: a family, not one magic dtype

Stage	Coord. 1	Coord. 2	Coord. 3	Coord. 4
Original b	0.25	-1.50	3.00	-0.75

FP8: a family, not one magic dtype

Stage	Coord. 1	Coord. 2	Coord. 3	Coord. 4
Original b	0.25	-1.50	3.00	-0.75
Payload P_b ($s = 0.75$)	0	-2	4	-1

FP8: a family, not one magic dtype

Stage	Coord. 1	Coord. 2	Coord. 3	Coord. 4
Original b	0.25	-1.50	3.00	-0.75
Payload P_b ($s = 0.75$)	0	-2	4	-1
Reconstruction $\widetilde{b}$	0	-1.50	3.00	-0.75

FP8: a family, not one magic dtype

Stage	Coord. 1	Coord. 2	Coord. 3	Coord. 4
Original b	0.25	-1.50	3.00	-0.75
Payload P_b ($s = 0.75$)	0	-2	4	-1
Reconstruction $\widetilde{b}$	0	-1.50	3.00	-0.75

- $E4M3$ spends more of its 8 bits on local spacing; $E5M2$ spends more on exponent range — "FP8" alone never fixes that tradeoff.
- **Scale-quantized scalar** (Def.): $p = Q_s(x) \in \arg\min_{q \in F_q} |x - sq|$, reconstructed as $\widetilde{x} = sp$.

FP8: a family, not one magic dtype

Stage	Coord. 1	Coord. 2	Coord. 3	Coord. 4
Original b	0.25	-1.50	3.00	-0.75
Payload P_b ($s = 0.75$)	0	-2	4	-1
Reconstruction $\widetilde{b}$	0	-1.50	3.00	-0.75

- $E4M3$ spends more of its 8 bits on local spacing; $E5M2$ spends more on exponent range — "FP8" alone never fixes that tradeoff.
- **Scale-quantized scalar** (Def.): $p = Q_s(x) \in \arg\min_{q \in F_q} |x - sq|$, reconstructed as $\widetilde{x} = sp$.

Coordinate 1 loses 0.25 to rounding here — the other three land exactly on this scale's grid. A coarser scale would lose more.

FP4: the same lifecycle, a much sharper tradeoff

Same block $b = (0.25, -1.50, 3.00, -0.75)$, same scale $s = 0.75$, now with the 5-code teaching set $\{-2, -1, 0, 1, 2\}$:

$$P_b = (0, -2, 2, -1), \qquad \widetilde{b} = (0, -1.50, 1.50, -0.75).$$

- The 3.00 coordinate clips to 1.50 — an error of 1.50, far larger than any FP8 rounding above.

FP4: the same lifecycle, a much sharper tradeoff

Same block $b = (0.25, -1.50, 3.00, -0.75)$, same scale $s = 0.75$, now with the 5-code teaching set $\{-2, -1, 0, 1, 2\}$:

$$P_b = (0, -2, 2, -1), \qquad \widetilde{b} = (0, -1.50, 1.50, -0.75).$$

- The 3.00 coordinate clips to 1.50 — an error of 1.50, far larger than any FP8 rounding above.
- An **FP4-style numerical path** (Def.) is five separate choices: payload bits, block layout, scale metadata and reconstruction, accumulation dtype, and output conversion — the payload alone does not determine the represented tensor.

FP4: the same lifecycle, a much sharper tradeoff

Same block $b = (0.25, -1.50, 3.00, -0.75)$, same scale $s = 0.75$, now with the 5-code teaching set $\{-2, -1, 0, 1, 2\}$:

$$P_b = (0, -2, 2, -1), \qquad \widetilde{b} = (0, -1.50, 1.50, -0.75).$$

- The 3.00 coordinate clips to 1.50 — an error of 1.50, far larger than any FP8 rounding above.
- An **FP4-style numerical path** (Def.) is five separate choices: payload bits, block layout, scale metadata and reconstruction, accumulation dtype, and output conversion — the payload alone does not determine the represented tensor.

Four bits force a sharper choice between covering the large value and resolving the small ones. NVFP4 and UE8M0 name source-defined pieces of this same picture [REPORTED: NVIDIA PTX documentation].

Granularity, accumulation, and propagation

Scale granularity is a partition, not a size

Granularity	Blocks (for X)	Scales	Tradeoff
Per tensor	all 32 entries	1	least metadata

Scale granularity is a partition, not a size

Granularity	Blocks (for X)	Scales	Tradeoff
Per tensor	all 32 entries	1	least metadata
Per row	4 rows	4	row adaptation

Scale granularity is a partition, not a size

Granularity	Blocks (for X)	Scales	Tradeoff
Per tensor	all 32 entries	1	least metadata
Per row	4 rows	4	row adaptation
Row block, $g = 4$	8 blocks	8	local adaptation

Scale granularity is a partition, not a size

Granularity	Blocks (for X)	Scales	Tradeoff
Per tensor	all 32 entries	1	least metadata
Per row	4 rows	4	row adaptation
Row block, $g = 4$	8 blocks	8	local adaptation

- DeepSeek-V3's report: 1×128 activation tiles, 128×128 weight blocks, partial sums promoted to a higher-precision accumulator every $N_C = 128$ terms [REPORTED: DeepSeek-V3 technical report].
- That is the training design as reported — DeepGEMM kernels and any toy fixture use their own scale rule.

Scale granularity is a partition, not a size

Granularity	Blocks (for X)	Scales	Tradeoff
Per tensor	all 32 entries	1	least metadata
Per row	4 rows	4	row adaptation
Row block, $g = 4$	8 blocks	8	local adaptation

- DeepSeek-V3's report: 1×128 activation tiles, 128×128 weight blocks, partial sums promoted to a higher-precision accumulator every $N_C = 128$ terms [REPORTED: DeepSeek-V3 technical report].
- That is the training design as reported — DeepGEMM kernels and any toy fixture use their own scale rule.

Same payload dtype, different scale partition, different reconstruction error — granularity is a numerical choice, not bookkeeping.

From concept to code: an FP8 KV-cache with per-tile scales

from `FlashMLA/tests/quant.py`

```
def quantize_k_cache(
    input_k_cache: torch.Tensor,    # (num_blocks, block_size, h_k, d)
    kvcache_layout: FP8KVCacheLayout,
) -> torch.Tensor:
    d, d_nope, d_rope, tile_size, num_tiles = kvcache_layout.get_meta()
    ...
```

From concept to code: an FP8 KV-cache with per-tile scales

from `FlashMLA/tests/quant.py`

```
def quantize_k_cache(
    input_k_cache: torch.Tensor,    # (num_blocks, block_size, h_k, d)
    kvcache_layout: FP8KVCacheLayout,
) -> torch.Tensor:
    d, d_nope, d_rope, tile_size, num_tiles = kvcache_layout.get_meta()
    ...
```

One helper, one real cache layout: per-tile FP8 scales for the non-RoPE portion, the RoPE portion kept in a documented higher-precision slice — exactly the granularity-plus-layout pairing this chapter keeps insisting on. [INSPECTED: FlashMLA `tests/quant.py`]

Where numerical error becomes a decision

The same error, two very different consequences

Wide margin

Logits $(5, 1, 0)$: a 0.05 perturbation barely moves the dominant softmax probability.

Near tie

Logits $(1.00, 0.99, 0)$: the same 0.05 perturbation can make the second logit the largest.

- Softmax's Jacobian has maximum row sum $2a_i(1 - a_i) \leq \frac{1}{2}$, so $\|\text{softmax}(z + e) - \text{softmax}(z)\|_\infty \leq \frac{1}{2} \|e\|_\infty$.

The same error, two very different consequences

Wide margin

Logits $(5, 1, 0)$: a 0.05 perturbation barely moves the dominant softmax probability.

Near tie

Logits $(1.00, 0.99, 0)$: the same 0.05 perturbation can make the second logit the largest.

- Softmax's Jacobian has maximum row sum $2a_i(1 - a_i) \leq \frac{1}{2}$, so $\|\text{softmax}(z + e) - \text{softmax}(z)\|_\infty \leq \frac{1}{2}\|e\|_\infty$.
- Router top-k selection is stable whenever the margin between the kth selected and largest unselected logit exceeds $2\|e\|_\infty$.

The same error, two very different consequences

Wide margin

Logits $(5, 1, 0)$: a 0.05 perturbation barely moves the dominant softmax probability.

Near tie

Logits $(1.00, 0.99, 0)$: the same 0.05 perturbation can make the second logit the largest.

- Softmax's Jacobian has maximum row sum $2a_i(1 - a_i) \leq \frac{1}{2}$, so $\|\text{softmax}(z + e) - \text{softmax}(z)\|_\infty \leq \frac{1}{2}\|e\|_\infty$.
- Router top-k selection is stable whenever the margin between the kth selected and largest unselected logit exceeds $2\|e\|_\infty$.

An aggregate tensor norm can pass while a near-tie flips the largest attention probability or the selected expert set.

Testing, training vs. inference, and what carries forward

Two different correctness problems

Inference

- Fixed weights, fixed model.
- Question: does the serving path preserve behavior for recorded prompts and shapes?

Training

- Errors can reach gradients, optimizer state, and many future updates.
- Question: does the quantized forward path change the trajectory?

Two different correctness problems

Inference

- Fixed weights, fixed model.
- Question: does the serving path preserve behavior for recorded prompts and shapes?

Training

- Errors can reach gradients, optimizer state, and many future updates.
- Question: does the quantized forward path change the trajectory?

A dtype path acceptable for inference is not automatically acceptable for training — the correctness evidence required for one does not transfer to the other.

What the retained H200 evidence actually closes

Surface	Retained result
Dense DeepGEMM	20 paired BF16/SM90-FP8 fixtures pass; worst `calc_diff` is 7.10×10^{-4} for FP8

What the retained H200 evidence actually closes

Surface	Retained result
Dense DeepGEMM	20 paired BF16/SM90-FP8 fixtures pass; worst `calc_diff` is 7.10×10^{-4} for FP8
FlashMLA FP8 KV cache	7 cases pass output, LSE, and RoPE-preservation gates

What the retained H200 evidence actually closes

Surface	Retained result
Dense DeepGEMM	20 paired BF16/SM90-FP8 fixtures pass; worst `calc_diff` is 7.10×10^{-4} for FP8
FlashMLA FP8 KV cache	7 cases pass output, LSE, and RoPE-preservation gates
MoE selected sets	12,636 rows; no declared margin condition is violated

What the retained H200 evidence actually closes

Surface	Retained result
Dense DeepGEMM	20 paired BF16/SM90-FP8 fixtures pass; worst `calc_diff` is 7.10×10^{-4} for FP8
FlashMLA FP8 KV cache	7 cases pass output, LSE, and RoPE-preservation gates
MoE selected sets	12,636 rows; no declared margin condition is violated

None of this defines a dtype-wide tolerance envelope. FP4 execution and other scale recipes remain outside the completed packet [OBSERVED: expanded H200 precision profiler packet].

Concept-to-code map for the chapter

- **scale-quantized scalar and payload/scale trace** → `code/chapter_precision_quantization_numerical_correctness/format_quant_reference.py`

Concept-to-code map for the chapter

- **scale-quantized scalar and payload/scale trace** → `code/chapter_precision_quantization_numerical_correctness/format_quant_reference.py`
- **FP8/FP4 kernel test suite and difference metric** → `DeepGEMM/tests/test_fp8_fp4.py`

Concept-to-code map for the chapter

- **scale-quantized scalar and payload/scale trace** → `code/chapter_precision_quantization_numerical_correctness/format_quant_reference.py`
- **FP8/FP4 kernel test suite and difference metric** → `DeepGEMM/tests/test_fp8_fp4.py`
- **scale-layout transform for a kernel's expected shape** → `DeepGEMM/csrc/apis/layout.hpp`

Concept-to-code map for the chapter

- **scale-quantized scalar and payload/scale trace** → `code/chapter_precision_quantization_numerical_correctness/format_quant_reference.py`
- **FP8/FP4 kernel test suite and difference metric** → `DeepGEMM/tests/test_fp8_fp4.py`
- **scale-layout transform for a kernel's expected shape** → `DeepGEMM/csrc/apis/layout.hpp`
- **FP8 KV-cache quantize/dequantize with per-tile scales** → `FlashMLA/tests/quant.py`

Concept-to-code map for the chapter

- **scale-quantized scalar and payload/scale trace** → `code/chapter_precision_quantization_numerical_correctness/format_quant_reference.py`
- **FP8/FP4 kernel test suite and difference metric** → `DeepGEMM/tests/test_fp8_fp4.py`
- **scale-layout transform for a kernel's expected shape** → `DeepGEMM/csrc/apis/layout.hpp`
- **FP8 KV-cache quantize/dequantize with per-tile scales** → `FlashMLA/tests/quant.py`
- **released 671B artifact's declared dtype** → `DeepSeek-V3/inference/configs/config_671B.json`

Concept-to-code map for the chapter

- **scale-quantized scalar and payload/scale trace** → `code/chapter_precision_quantization_numerical_correctness/format_quant_reference.py`
- **FP8/FP4 kernel test suite and difference metric** → `DeepGEMM/tests/test_fp8_fp4.py`
- **scale-layout transform for a kernel's expected shape** → `DeepGEMM/csrc/apis/layout.hpp`
- **FP8 KV-cache quantize/dequantize with per-tile scales** → `FlashMLA/tests/quant.py`
- **released 671B artifact's declared dtype** → `DeepSeek-V3/inference/configs/config_671B.json`

Name the reference, metric, and tolerance before trusting a small payload.

Next: twelve sections work through this contract in depth — from the running example's invariants to the correctness-test report that closes the chapter.